1. Why design matters

- Messy tables = repeated data and mistakes
- Good design: each fact stored once
- Normalization = simple tidy-table rules

2. 1NF, 2NF, 3NF in plain words

- 1NF: one value per cell (no lists)
- 2NF: depend on the WHOLE key
- 3NF: no middle-man columns (city to state)

```
SELECT m.name, c.state FROM members AS m
JOIN cities AS c ON m.city = c.city;
```

name	state
Aisha	Texas
Ben	Colorado
Carlos	Texas
Dana	Massachusetts
Eli	Colorado

3. Relationships

- One-to-one: member and library card
- One-to-many: FK on the many side
- Many-to-many: needs a junction table

4. ER diagrams

- Boxes = tables, with their columns
- Lines = relationships (1 to many)
- Free tools: draw.io, dbdiagram.io

5. Practice free + what's next

- sqliteonline.com, db-fiddle.com, sqlime.org (choose SQLite)
- sqlbolt.com for free lessons
- Paste the setup from the practice guide, click Run
- Next, Part 10: final project, a pet shop database